

FURNACE v5.3.1 — Response to the verified review

Result: all 28 reviewer mutations now rejected (was 0/28). All 9 positive controls still fire. CREATE3 re-derivation reproduces all 9 registry addresses in-validator. The Conviction oracle reproduces the review's §6 numbers to 1e-9. 66/66 checks pass (`furnace.review.test.ts`). **No monetary parameter changed.**

v5.3.0 hash `0x1554...94ed5` reproduced exactly (26,829 chars / 26,875 UTF-8 bytes — review §3 confirmed).

v5.3.1 hash:

`0x9ecedb63d5d10951caaf506032a1cf6d14b9e2196f236e2d2a45c200870463f7`

Disposition

§	Finding	Status
4.1	Two withdrawal helpers disagree	Fixed — legacy delegates to cap-first helper
4.1	<code>deedHeatAge(730)</code> = 584	Fixed — caps before carry → 292
4.2	<code>cancelReturnsAsFreshLot</code> , lapse toast, casting copy vs resumeFrozen	Fixed + validator guard
4.3	Casting LP into single-token pools	Fixed conservatively (liquid EMBER) — see Decision 2

§	Finding	Status
4.4	Cast LP vs burned match LP ownership	Fixed — user's cast share → source lot; protocol's pairing → burned
5	\$87.30 is quote-side only; mark is ~\$174.60	Fixed in spec §3/§26 and config
6	Conviction cap $b \times \text{earnings}$ overpays h-fold	Fixed — additive $L \cdot s \cdot b / W$; <code>epochAllocation()</code> oracle added
6	Empty bucket makes split 27/73	Documented — see Decision 1
7	28 undetected mutations	Fixed — 28/28
7	19-byte Ethereum router	Fixed — Uniswap's published Router02, EIP-55 verified
7	Missing guardians undetected	Fixed — every enabled chain, VM-specific encoding
8	Mint helpers accept negatives	Fixed — <code>RangeError</code> on negative/non-finite
8	Pyre 25% "no rebate" claim	Fixed wording — reduces, not eliminates
8	Global mint ledger, OFT in-flight, funded refunds, bounded accrual	Open — contract work, not config
—	New: treasury addresses were decorative fakes	Fixed — explicitly unset + validated

The new finding

`treasury.addresses` held
`0x7Ea596B12cD34Ef56aB78Cd90Ef12Ab34Cd56F1E` (fails EIP-55) and
`F1REtreasury596aBcDeFGHiJkLmNoPqRsTuVwXyZ123` (valid 32-byte
base58, but hand-typed — no one holds the key). Every 3% genesis fee
routes there. A zero address looks wrong in a deploy log; these didn't.
They're now explicit blanks that block the build. The deploy script must
assert they equal the deployed Safe / Squads vault.

Correction to my own work

My first pass "fixed" the 19-byte router by padding it to `...c05e9` — the
guess the review told me not to make. It fails EIP-55. Replaced with the
address from Uniswap's deployments page, and the checksum sweep
now makes that whole class of error a build failure.

Decisions for you (not silently made)

1. Where an unused Conviction bucket returns. Spec: Heat pool only →
27/73 in no-seal epochs. Alternative: pro-rata → 30/70 holds in every
epoch.

Recommendation (moderate confidence): "30% newcomer
floor" is a headline promise and a validated identity number. Heat-only
quietly shrinks it whenever nobody is sealed. The cost is 3pp to
veterans in those epochs. Counter-argument: unused loyalty budget
going to the loyalty pool is coherent. Either is defensible; the label must
match the choice.

2. Casting in Ember Vault. I applied liquid EMBER to both single-token
pools. That's overbroad: Ember Vault holds EMBER, so the cast share
can join the source lot as *EMBER* with no pairing at all. That keeps "25%
is never paid liquid" true there.

Recommendation (high confidence): per-pool rule — LP pools → source-
lot LP; Ember Vault → source-lot EMBER; Cold Storage → liquid EMBER
(it's the base-emissions-only pool by design).

3. Contract-layer items (§8) — authenticated global supply ledger
counting in-flight OFT, funded Kindling refunds, bounded accrual,

integer settlement with explicit residuals. These belong in the Foundry/Anchor invariant suite.

Dependency

The validator now imports `@noble/hashes` (zero-dependency, audited) for Keccak-256. `pnpm add @noble/hashes`.